

PRO CLASSROOM

Project Synopsis

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MASTERS OF COMPUTER APPLICATION

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Introduction

Nowadays, schools and colleges use online tools so students can easily turn in their assignments. However, manually checking for copied work means teachers have to open each PDF and compare assignments one by one, which takes a lot of time and can lead to mistakes.

With our system, we extract text (even from scanned pages) and automatically compare all submitted PDFs. Instructors then receive a clear similarity percentage for each pair of assignments.

This approach saves teachers a significant amount of time and ensures any copied content is spotted promptly.

Background

In many online learning environments, students submit assignments as PDF files, but educators still rely on manual checks to catch copied work. This process not only demands significant time — opening, reading, and comparing each document — but also risks missing similarities, especially when students paraphrase content.

Existing systems focus on basic file management and deadlines, but they often lack built-in tools for similarity detection or force instructors to use separate workflows. Our project addresses this gap by embedding an automated similarity check directly into the submission platform, creating a seamless experience for both students and instructors.

Objectives

- Provide a simple upload portal for PDF assignments.
- Automatically extract and process text from both printed and handwritten documents.
- Compute pairwise similarity scores and present clear percentage-based reports.
- Highlight matching text segments to aid instructor review.
- Offer configurable similarity thresholds and batch processing options.

Purpose

The primary purpose of this system is to streamline the plagiarism detection process for educators. By integrating similarity analysis directly into the submission workflow, we aim to reduce manual workload, increase review accuracy, and maintain high standards of academic integrity.

Scope

- PDF assignments only (no support for other formats).
- Text-based similarity detection, including OCR for scanned or handwritten content.
- English-language content focus.
- Instructor dashboard for individual and batch reviews.
- No code plagiarism or multimedia comparison in this phase.

Applicability

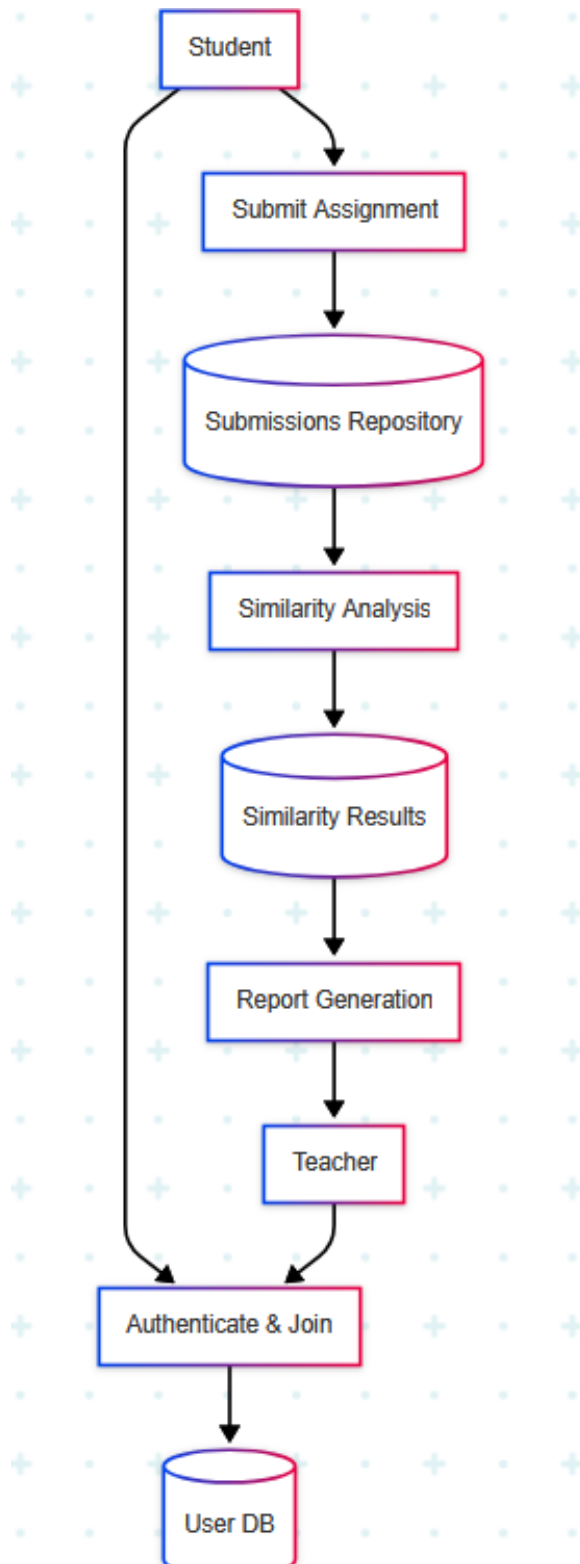
This system is designed for educational institutions—from schools to universities—seeking a built-in solution for assignment submission and plagiarism detection. It can be integrated as a standalone tool or as part of an existing Learning Management System (LMS).

Tech Stack

- **Front-End:** React.js for a responsive upload and review interface.
- **Back-End:** Node.js (Express) for API endpoints, with a Python microservice handling PDF text extraction and similarity comparisons.
- **Database:** MongoDB for flexible storage of user data, submissions, and similarity results.
- **OCR Engine:** Lama Vision model for detecting and extracting both printed and handwritten text from PDFs.
- **Similarity Algorithm:** Python's built-in difflib library for sequence matching, supplemented by TF-IDF vectorization and cosine similarity in the microservice.

Design Diagrams

DFD



Modules**

1. Teacher Module

- The teacher logs into the system, creates and manages classrooms, and sets up assignments with deadlines.
- Teachers can view the number of student submissions for each assignment and access detailed similarity reports.

2. Student Module

- Students register or log in, then join classrooms using a unique code provided by the teacher.
- Once enrolled, students can view active assignments and upload their PDF submissions directly through the portal.

3. Assignment Checking Module

- This module automatically compares two selected PDF submissions, computing similarity percentages.
- It highlights matching text segments side by side, enabling quick review of potential overlaps.

Future Enhancements

- Expand file format compatibility to include Microsoft Word (.docx) and PowerPoint (.pptx), enabling similarity checks on a wider range of assignments.
- Incorporate code plagiarism detection for programming coursework, using syntax-aware comparison algorithms to identify copied or similar code segments.
- Implement real-time notifications to promptly alert instructors when submission pairs exceed defined similarity thresholds.
- Develop analytics dashboards offering class-wide insights into submission trends, similarity distributions, and student performance metrics.
- Plan integration with popular Learning Management Systems (e.g., Google Classroom, Moodle) to provide seamless assignment management and single sign-on.